

# ARYAN VASHISHTH

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## Summary

Software Engineer building real-time backends for **AI-powered products**. At Stimuler, shipped Conversation V3 end-to-end (**94% p50 latency reduction**), real-time voice calling on **Gemini Live**, and **multi-LLM orchestration** integrating 4+ providers. Experienced in distributed systems, TypeScript/Node.js, real-time architectures, and building reliable systems from concept to production.

## Education

**Thapar Institute of Engineering and Technology**  
BE, Computer Engineering, CGPA: 8.32/10

May 2022 – May 2026  
Patiala, Punjab

## Skills

**Languages:** C++, Java, JavaScript, TypeScript, SQL, Dart  
**Computer Science:** Data Structures & Algorithms, Object-Oriented Design, System Design, Operating Systems, DBMS, Distributed Systems, Computer Networks  
**Backend:** Node.js, Bun, Hono, Express.js, REST APIs, WebSockets, Microservices, Caching, Redis  
**Frontend/Mobile:** React.js, Next.js, Flutter, Kotlin, Kotlin Multiplatform, XML, React Native  
**Databases:** PostgreSQL, MongoDB, Firestore, Redis  
**Cloud & DevOps:** AWS (EC2, ECS, Lambda, S3), Firebase, Docker, Nginx, CI/CD, Git  
**Certifications:** AWS Certified AI Practitioner (AIF-C01); Blockchain by IIT Kanpur

## Experience

### MOVE IN SYNC | Software Engineer

Aug 2026 – Present

- Migrating the Field App codebase with 40+ modules from Flutter to Kotlin Multiplatform, standardizing the stack around shared business logic and native Android components.
- Built an **AI intelligence layer for production alerting**, reducing **alert noise by 83%**. Solved Crashlytics issues, improving the application's **crash-free rate from 99.6% to 99.8%**.
- Led **VAPT security fixes** across the mobile applications, fixing **30+ vulnerabilities**, while developing a highly **configurable Security Sign-off workflow** across Office and Field applications.

### BASETHESIS LABS | Software Engineer

Jun 2026 – Jul 2026

- Owned **Synth's TallyPrime integration** for an **AI CFO (76 users, kayess.chat.getsynth.io)**. Unlike Zoho/Microsoft/ERPNext, **Tally has no API**; reverse-engineered its XML protocol.
- Built a **deterministic financial-statement engine** (P&L, BS, TB, aging, TDS, GST) **matched Tally** for 15 clients.
- Turned Tally's flaky single-threaded XML gateway into a fault-tolerant production write path with a **half-open circuit breaker**, single-flight mutex, and **month-windowed batched reads** (byte + wall-clock caps); root-caused and eliminated a reproducible **OOM wedge** from unbounded full-history exports.

### STIMULER | Software Engineer Intern

Jan 2026 – Jun 2026

- Cut Conversation V3 p50 latency **23s → ~1.4s (~94%)** by re-architecting backend off Firebase Cloud Functions onto **Bun + Hono** for native WebSocket streaming; drove the system to upstream-bound via 2 cross-team optimizations.
- Rebuilt voice calling server-side on **Gemini Live + Bun + Hono** (client-side Hume couldn't capture audio); shipped **24kHz PCM streaming**, 500ms barge-in stitching, **dynamic RMS noise gating**; killed Gemini's post-2-min latency cliff; rolled out to **~30K users** via SHA-1 bucketing.
- Architected V3's **multi-LLM runtime** with **4+ providers** (OpenAI, GPT-4o, in-house on **Modal + vLLM**), swappable live via Firestore config; engineered **Qwen3 output gates** blocking regressions without latency cost.
- Built calling-ops dashboard solo (**Flutter Web**, Firebase Auth); shipped a **cross-session memory layer** that LLM-summarizes voice + chat into per-user profiles threaded into every prompt.
- Shipped **self-healing V2→V3 migration** (lazy, per-user) across **150+ topics**; zero downtime, **40+ Jest tests**.

## Projects

### Distributed Trading Engine | TypeScript, Node.js, Redis | [GitHub](#)

- Built a heap-based order book (**O(log n)** insertion + best-price matching) and an event-driven matching engine on **Redis Pub/Sub**; sustained **10,000+** orders with partial fills and real-time trade ID generation.
- Built an async order-processing pipeline with execution lifecycle tracking (open → matched → filled); **non-blocking event loop** maintained consistent match latency under concurrent order submissions.

### ShopXar | React.js, TypeScript, Node.js, MongoDB, Three.js, AR/3D | [Live Demo](#)

- Built a **3D e-commerce platform with AR previews** (Three.js, model-viewer); cut model-load latency **40%** via CDN caching, achieved **100% cross-device compatibility**.